N7 — The Single Open Step: The Riemann Hypothesis as the Even-Sector Convergence of the Finite Weil Form

Concretely, the even moments $M_{2k} = \int \mathcal{S}_g^{2k}\eta \geq 0$ are definite but vacuous (true for every zero configuration,

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Abstract

We record the terminal status of the apex/cubic program's assault on the Riemann Hypothesis, integrated with Connes' prolate/Weil-positivity strategy. The Hypothesis is reduced — exactly, and with the finite half realized unconditionally — to a single analytic step: the even-sector convergence of the finite-prime Weil quadratic form's minimal eigenvector to Riemann's Ξ. We prove or certify every link except that step, and we show (adversarially, across six independent routes) that each of those routes closes at the same point, so none reaches the step from the program's available structures without circularity. We state the open terminus precisely, separate it cleanly from what is established, and decline — by the program's own discipline — to represent it as proved. As of Connes–Consani–Moscovici (November 2025), this step is open in the field; we add to it a from-scratch finite realization and an adversarially-verified map of why the routes around it close. Section 5 sharpens the step. Because the finite approximants are real-rooted, the Hadamard product controls their growth by one scalar, the log-variance $S(\lambda)$, so convergence is needed only on one real segment below the first zero, and the prolate proximity step needs only a logarithmic rate; we prove that a finite set of off-line zeros would force $S(\lambda) \to \infty$. Running the identical certified machinery on the Davenport–Heilbronn function, which shares the functional equation but has no Euler product, shows that evenness of the ground state carries no Riemann-Hypothesis content, while boundedness of S does: off the Euler product the ground state leaves the bulk for the window edges.

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1. The reduction chain

The Riemann Hypothesis is equivalent to each of the following, the equivalences being unconditional:

$$\text{RH} \iff E_B = \Xi + i\Xi' \text{ Hermite–Biehler} \iff m_B = -\Xi'/\Xi \text{ Herglotz} \iff \varphi'(x) \geq 0 \text{ (monotone clock)} \iff \operatorname{Re}\mathcal{C}_\zeta \geq 0 \,($$

Connes' route adds a finite-to-infinite bridge. The finite-prime Weil quadratic form on test functions supported in $[\lambda^{-1}, \lambda]$ (primes $\leq \lambda^2$),

$$W_\lambda(\varphi) = 2\Big(\int \varphi\, e^{x/2}\Big)\Big(\int \varphi\, e^{-x/2}\Big) + \frac{1}{2\pi}\int |\widehat{\varphi}(r)|^2\, \Omega_\infty(r)\, dr - 2\sum_{n\leq\lambda^2} \frac{\Lambda(n)}{\sqrt{n}}(\varphi\star\varphi)(\log n), \quad \Omega_\infty(r) = \operatorname{Re}\psi(\tfrac{1}{4}+\tfrac{ir}{2}) - \log\pi,$$

has a minimal eigenvector θ_λ whose Fourier zeros lie **on the critical line** (Connes Thm 6.1; Carathéodory–Fejér, 1911). Riemann's identity gives the target: $\widehat{\mathcal{E}(h)} = \Xi$ with $\mathcal{E}(f)(u) = u^{1/2}\sum_{n\geq 1} f(nu)$, $h(u) = \frac{\pi}{2}u^2(2\pi u^2 - 3)e^{-\pi u^2}$. **The single open step is the convergence** $\theta_\lambda \to k = \mathcal{E}(h)$; Hurwitz's theorem then forces Ξ's zeros onto the line. Connes–Consani–Moscovici (2025) state plainly that a rigorous proof of this convergence would establish RH.

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2. The finite half is realized (unconditional) [P]

Theorem 2.1 (finite Weil form generates the zeros). *Built from the closed-form archimedean kernel $K_\infty(u) = -e^{-|u|/2}/(1-e^{-2|u|})$, the rank-two pole term, and the prime corrections for primes $\leq X$, the finite Weil form's even-sector minimal eigenvector has Fourier zeros matching the first eight nontrivial zeta zeros to < 0.2* (certified, `probe_weil_form_v3.py`: $14.13, 21.02, 25.01, 30.42, 32.93, \ldots$, 8/8). The convergence target is confirmed independently: $\widehat{\mathcal{E}(h)}$ has zeros at the zeta zeros to 5×10^{-3} (`probe_convergence_target.py`).

Theorem 2.2 (prolate ground state, the vehicle). *The prolate operator $PW_\lambda = -\partial_x((\lambda^2 - x^2)\partial_x) + (2\pi\lambda x)^2$ has a simple, even ground state with $O(\lambda^2)$ spectral gap and parity alternating E, O, E, O* (certified, `probe_prolate_simple_even.py`). This is the "known for the prolate" half of Connes [C1, Sect. 6.6(a)].

Thus both ends are in hand unconditionally: each finite truncation has its zeros on the line, and the limit target k has transform Ξ.

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3. Six routes, one closure point [A]

An exhaustive adversarial analysis (six independent routes, each refuted by an independent skeptic checking for a hidden RH assumption) establishes:

Proposition 3.1 (common closure point). *Every route to the convergence step reduces, at its closure point, to the same archimedean limit positivity $\varphi' \geq 0 \iff$ RH. None yields a non-circular advance.* The routes and their sharp closure points:

route	closes at	status
prolate-perturbation	prolate gap $\sim \lambda^2$ **grows**, Weil gap $\sim e^{-4\pi e^L}$ **shrinks**; perturbation = gap	ruled out in operator norm; the trial-vector form is live (Corollary 5.5)
cubic uniform rigidity	witness margin $\sim \gamma_0^{-3}$ collapses; no uniform-in-height bound	ruled out
de Branges canonical system	$H(t) = P_{\varphi(t)}$ positive only because t valid $\iff \varphi' \geq 0 \iff$ RH	circular
function-field bridge	apex form is **marginal** ($\lambda_{\min} \to 0$); no $\mathbb{F}_1$ surface, no finite-rank Hodge positivity	circular
Arakelov intersection	local forcing $\lvert J_p\rvert = \sqrt{p}$ unconditional, but global assembly = GRH; Hodge index on finite-rank (rank-0) lattice cannot host the zeros	circular
Beurling–Nyman	closure **rate** is Baez-Duarte-equivalent to RH	circular
cumulant / moment tower	every order: definite ⇒ vacuous, contentful ⇒ indefinite (Prop 3.2)	ruled out

The two load-bearing equivalences are recorded in the program's own files (`canonical_system_phase.md` Part 3: the entire content of the canonical-system positivity is $\varphi' \geq 0$; `Note_BuresPickGram`: the positive Gram law "is equivalent to RH"). The $\mathbb{Z}/6$-fiber candidate $L(s, \chi_3)/\Xi$ is a proven **non-Herglotz** no-go (`Note_a9Q7x`).

The last two rows are recent and sharpen the picture from "no route found" to a structural reason, so we state them as theorems rather than route-summaries.

Proposition 3.2 (the order-tower dichotomy). *Let $\mathcal{S}_g(x) = \sum_\gamma \tilde{g}(\frac{1}{2} + i\gamma)e^{i\gamma x}$ be the zero-spectrum sum and consider, for each order $m \geq 2$, the diagonal moment $M_m = \int \mathcal{S}_g^m \, \eta$ and the connected cumulant κ_m (its Wick-reduced part). Over $\mathbb{Q}$, at every order $m \geq 2$, definiteness and arithmetic content are mutually exclusive:*

$$\text{(sign-definite for all } g) \iff \text{(RH-independent, i.e. vacuous)}, \qquad \text{(carries RH content)} \iff \text{(indefinite)}.$$

Concretely, the even moments $M_{2k} = \int \mathcal{S}_g^{2k} \eta \geq 0$ are definite but vacuous (true for every zero configuration, RH or not — verified for M_4 on both an on-line and an off-line configuration); the odd moments M_{2k+1} are indefinite by parity ($\mathcal{S}_g \mapsto -\mathcal{S}_g$); and the connected even cumulants are indefinite because the Wick subtraction removes exactly the Gaussian part that supplied positivity. For $m = 4$ the connected cumulant is the certified selection rule

$$\kappa_4(\chi_a, \chi_b, \chi_c, \chi_d)\big|_u = \delta_{a+b+c+d\equiv 0} - \delta_{a+b\equiv 0}\delta_{c+d\equiv 0} - \delta_{a+c\equiv 0}\delta_{b+d\equiv 0} - \delta_{a+d\equiv 0}\delta_{b+c\equiv 0},$$

which takes values $\{+1, 0, -1, -2\}$ *on* $\mathbb{Z}/N$ *— indefinite.* [P] (`probe_even_order_forcing_dichotomy.py`, cross-checking the $\mathbb{Z}/30$ counts 1082/120/914 of the quartic selection paper).

The single exception is order two **over** $\mathbb{F}_q$, where the form is the Hodge/intersection pairing on a genuine surface and definiteness (Hodge index) coincides with RH-content (the Frobenius eigenvalue moduli $|\alpha| = \sqrt{q}$, equivalently the unconditional Eisenstein norm $a^2 - ab + b^2 = p$ of the apex/Jacobi-sum amplitude). The dichotomy thus localizes the entire difficulty to one place: the construction, at the archimedean place over $\mathbb{Q}$, of a single object that is — like the $\mathbb{F}_q$ Hodge form, and unlike every arithmetic order — both definite and RH-encoding. This is precisely the apex–archimedean operator of N4–N6, whose order-two Weil form is known to be **marginally** positive; RH is the statement that this lone candidate-exception to Proposition 3.2 is positive rather than merely marginal [C].

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4. The terminus: the even–odd split $\Delta(\lambda)$ [C]

The simplicity-and-evenness precondition (Connes [C1, Sect. 6.6(a)]) for the limit transfer is the sign of the even–odd ground split

$$\Delta(\lambda) = \lambda_0^{\mathrm{odd}}(W_\lambda) - \lambda_0^{\mathrm{even}}(W_\lambda).$$

Observation 4.1 (the split is doubly-exponentially small in absolute terms, large in relative terms). *The ground eigenvalue tracks the first self-dual prolate defect,* $\varepsilon(\lambda) \approx c\,(1 - \chi_4(\lambda))$ *with* $c \approx 6$–12 *([N2], Prop. 4.3), so in absolute terms the split is doubly exponential in* $L = 2\log\lambda$. *It is nevertheless resolved directly by the closed-form matrix of Section 5:* $\varepsilon_{\mathrm{even}} = 9.2 \times 10^{-13}$ *against* $\varepsilon_{\mathrm{odd}} = 5.4 \times 10^{-10}$ *at* $\lambda = 2$, *and* 4.0×10^{-75} *against* 5.9×10^{-71} *at* $\lambda = 4$ *(Table 5.1).* The relative split grows along the ladder. The earlier probes (`probe_weil_form_v3.py`, `probe_weil_evenodd_split.py`) sat at a 10^{-5} regularization floor; their reading that the split lies "below any feasible numerical resolution" is withdrawn.

Observation 4.2 (the sign of the split is not the Riemann Hypothesis). *We had argued that the minimizer of* $\sum_\rho |\widehat{\varphi}(\gamma_\rho)|^2$ *wins by being* Ξ, *which is even, so that "even ground state"* $\iff$ *"the minimizer is* Ξ" $\iff$ *RH. The matched pair of Section 5.5 refutes the first equivalence. The Davenport–Heilbronn form, whose function violates RH, has a simple even ground state at every window past its transition, and its ground-state transforms then have only real zeros (Observation 5.8). Evenness is the parity of the dominant mode. It is necessary for the Carathéodory–Fejér mechanism, but it does not carry the Hypothesis.*

The open step, restated. *The single step is the conjunction of (a), simple even ground states along a sequence* $\lambda_j \to \infty$, *with the interval convergence of Corollary 5.3; equivalently (a) with the spectral inequality of Corollary 5.5.* Either conjunction implies RH (Section 5). The converse is open as well: the CCM convergence is not known even assuming RH. The program's earlier framing, in which the sign of Δ is carried by the first parity-asymmetric term anchored by $L(\frac{1}{2}, \chi_3) = 0.4809$, survives only as a heuristic for (a); the Hypothesis itself sits in the normality of the approximant family (Theorems 5.2 and 5.6).

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5. Laguerre–Pólya transport: the open step needs one interval, and the matched pair shows where it lives

Sections 2–4 locate the open step in the convergence $\widehat{\theta}_\lambda \to \Xi$. Connes–Consani–Moscovici [C2, Sect. 8] state the requirement as uniform convergence on closed substrips of $|\operatorname{Im} z| < \frac{1}{2}$, which Hurwitz's theorem then transports to the zeros. This section shows that the requirement is far stronger than needed, isolates the one scalar that carries the content, and runs the identical machinery on a function with the same functional equation but no Euler product, to see which hypothesis fails when the Riemann Hypothesis does.

Throughout, $\lambda > 1$, $L = 2\log\lambda$, QW_λ is the semilocal Weil form on $L^2([\lambda^{-1},\lambda], d^*u)$ (or its CCM truncation QW_λ^N on V_n, $|n| \le N$), ξ_λ its ground state, $\varepsilon(\lambda)$ its smallest eigenvalue, and $\widehat{f}(z) = \int f(u)u^{-iz}\,d^*u$. Hypothesis **(a)** is the first missing step of [C2, Sect. 8]: *the smallest eigenvalue is simple and its eigenvector even.* Under (a), $\widehat{\xi}_\lambda$ is an even entire function of exponential type whose zeros are all real ([CvS]; [C2, Thm 5.10(iii)] for QW_λ^N).

5.1 The transport theorem [P]

Let $\mathcal{L}$ be the class of even real entire functions F of genus at most one with only real zeros and $F(0) = 1$. By Hadamard, $F(z) = \prod_k (1 - z^2/a_k^2)$ over its positive zeros $a_1 \le a_2 \le \cdots$, and we write

$$S(F) \;=\; \sum_k a_k^{-2} \;=\; -\tfrac{1}{2}F''(0).$$

Lemma 5.1 (the Hadamard–Laguerre bounds). *For $F \in \mathcal{L}$ with $S = S(F)$: (i) $|F(z)| \le e^{S|z|^2}$ for all $z \in \mathbb{C}$; (ii) for $0 \le t < a_1$, $e^{-St^2/(1-t^2/a_1^2)} \le F(t) \le e^{-St^2}$.*

Proof. (i) $|1 - z^2/a^2| \le 1 + |z|^2/a^2 \le e^{|z|^2/a^2}$. (ii) For $x \in [0,1)$, $-x/(1-x) \le \log(1-x) \le -x$; apply with $x = t^2/a_k^2 \le t^2/a_1^2$ and sum. □

Theorem 5.2 (Laguerre–Pólya transport). *Let $F_j \in \mathcal{L}$ and suppose $F_j \to G$ uniformly on one real segment $[0,\eta]$, $\eta > 0$, with G continuous and positive there. Then:* 1. $\limsup_j S(F_j) \le -\log G(\eta)/\eta^2$; 2. *$\{F_j\}$ is a normal family on $\mathbb{C}$, and every locally uniform limit is an even entire function with only real zeros that agrees with G on $[0,\eta]$;* 3. *if G is the restriction of an entire function $\widehat{G}$, then $F_j \to \widehat{G}$ locally uniformly on $\mathbb{C}$, and $\widehat{G}$ has only real zeros.*

Proof. (1) Uniform convergence to a positive G makes $F_j > 0$ on $[0,\eta]$ for large j, so $a_1(F_j) > \eta$, and Lemma 5.1(ii) at $t = \eta$ gives $S(F_j) \le -\log F_j(\eta)/\eta^2 \to -\log G(\eta)/\eta^2$. (2) Lemma 5.1(i) makes the family locally bounded; Montel gives normality. A limit F satisfies $F(0) = 1$, so $F \not\equiv 0$, and by Hurwitz it has no zeros off $\mathbb{R}$, since the F_j have none. (3) Every subsequential limit equals $\widehat{G}$ on $[0,\eta]$, hence everywhere; so the whole sequence converges to $\widehat{G}$. □

The point of Theorem 5.2 is that for real-rooted families the *growth* control that Hurwitz needs is free: the single scalar S supplies it, and S is bounded as soon as the family converges at one real point below its first zero.

5.2 Consequences for the open step [P]

Corollary 5.3 (the second missing step, interval form). *Suppose that along a sequence $\lambda_j \to \infty$ hypothesis (a) holds, $\widehat{\xi}_{\lambda_j}(0) \ne 0$, and for some $\eta > 0$*

$$\frac{\widehat{\xi}_{\lambda_j}(t)}{\widehat{\xi}_{\lambda_j}(0)} \;\longrightarrow\; \frac{\Xi(t)}{\Xi(0)} \qquad \text{uniformly on } [0,\eta].$$

Then the Riemann Hypothesis holds. The same holds for the truncated CCM ground states $\xi_{\lambda,N(\lambda)}$.

Proof. Under (a), $F_j = \widehat{\xi}_{\lambda_j}/\widehat{\xi}_{\lambda_j}(0) \in \mathcal{L}$. Since $\Xi(t) = \int \Phi(u)\cos(tu)\,du$ with Riemann's kernel $\Phi > 0$, we have $\Xi(0) > 0$ and $\Xi > 0$ on $[0,\eta]$ for $\eta < \gamma_1$. Apply Theorem 5.2 with $\widehat{G} = \Xi/\Xi(0)$. □

Corollary 5.3 replaces "uniformly on closed substrips of $|\operatorname{Im} z| < \frac{1}{2}$" by "uniformly on one real segment, of any length, below the first zero". The segment carries no zeros of Ξ at all; the transport of the zeros is done by the real-rootedness of the approximants.

Corollary 5.4 (proximity at a logarithmic rate suffices). *Let $k_\lambda = \mathcal{E}(h_\lambda)$ be the prolate ansatz of [C2, Sect. 7], $\tilde{k}_\lambda$ its even part, and θ_λ the angle between ξ_λ and $\tilde{k}_\lambda$ in $L^2(d^*u)$. If (a) holds along $\lambda_j \to \infty$ and*

$$\sin\theta_{\lambda_j} \cdot \sqrt{\log \lambda_j} \;\longrightarrow\; 0,$$

then the Riemann Hypothesis holds.

Proof. Normalize $\|\xi\| = 1$ and write $\xi = c\,\tilde{k}/\|\tilde{k}\| + e$ with $e \perp \tilde{k}$, $\|e\| = \sin\theta$, $c = \cos\theta$. For real t, $|\hat{f}(t)| \le \|f\|_{L^1(d^*u)} \le \sqrt{L}\,\|f\|_{L^2}$, so

$$\Big|\frac{\hat{\xi}(t)}{\hat{\xi}(0)} - \frac{\hat{\tilde{k}}(t)}{\hat{\tilde{k}}(0)}\Big| = \frac{|\hat{e}(t)\hat{\tilde{k}}(0) - \hat{\tilde{k}}(t)\hat{e}(0)|}{|\hat{\xi}(0)\hat{\tilde{k}}(0)|} \le \frac{\sqrt{L}\,\sin\theta\,(1 + \sup_t |\hat{\tilde{k}}(t)|/|\hat{\tilde{k}}(0)|)}{|\hat{\xi}(0)|}.$$

By [C2, Lem. 7.3] (resting on the classical prolate estimate [C2, Lem. 7.2]), $\hat{k}_\lambda$, and hence $\hat{\tilde{k}}_\lambda$ since Ξ is even, converges to a fixed multiple of Ξ uniformly on closed substrips, in particular on all of $\mathbb{R}$; and $|\Xi(t)| \le \int \Phi = \Xi(0)$ on $\mathbb{R}$. So the supremum is bounded, $\hat{\tilde{k}}_\lambda(0)/\|\tilde{k}_\lambda\|$ is bounded away from 0, and $|\hat{\xi}(0)| \ge c\,|\hat{\tilde{k}}(0)|/\|\tilde{k}\| - \sqrt{L}\sin\theta$ is bounded away from 0 once $\sqrt{L}\sin\theta \to 0$. The right side then tends to 0 uniformly on $\mathbb{R}$, and Corollary 5.3 applies. □

Corollary 5.5 (the spectral form). *Let $\mu_1(\lambda)$ be the second eigenvalue of QW_λ on even functions and $R(f) = QW_\lambda(f)/\|f\|^2$. If (a) holds along $\lambda_j \to \infty$ and*

$$\frac{R(\tilde{k}_\lambda) - \varepsilon(\lambda)}{\mu_1(\lambda) - \varepsilon(\lambda)}\,\log\lambda \;\longrightarrow\; 0,$$

then the Riemann Hypothesis holds.

Proof. Write $\tilde{k}/\|\tilde{k}\| = \cos\theta\,\xi + \sin\theta\,w$ with w even, $w \perp \xi$, $\|w\| = 1$. Since ξ is an eigenvector, $R(\tilde{k}) = \cos^2\theta\,\varepsilon + \sin^2\theta\,R(w) \ge \cos^2\theta\,\varepsilon + \sin^2\theta\,\mu_1$, so $\sin^2\theta \le (R(\tilde{k}) - \varepsilon)/(\mu_1 - \varepsilon)$. Apply Corollary 5.4. □

Corollary 5.5 turns the second missing step into an inequality between two numbers of the semilocal Weil form: the Rayleigh excess of an explicit trial vector, and the second even eigenvalue. The required margin is only logarithmic.

The Rayleigh excess is controlled in closed form by the first self-dual prolate defect. The exact tail identity of the companion note [N2] (Lem. 3.2) gives $QW_\lambda(k_\lambda) = QW(F_{\text{out}})$, and $R(k_\lambda)/(1 - \chi_4(\lambda))$ stays between 4.9 and 13.2 for $\lambda \in [1.4, 1.7]$ ([N2], Sect. 3). Where the excess has been measured directly ($\lambda = 1.5$: $R(\tilde{k}) - \varepsilon = 2.12 \times 10^{-5}$ in [N2], with $\mu_1 - \varepsilon = 0.275$ from Table 5.1), the left side of Corollary 5.5 is 3×10^{-5}.

Beyond $\lambda = 1.7$ the finite-difference prolate of [N2] is not accurate enough to measure $R(\tilde{k}_\lambda)$. What Table 5.1 does certify at every rung is the denominator. The ratio μ_1/ε grows from 1.9×10^3 at $\lambda = 1.5$ to 1.2×10^8 at $\lambda = 4$. Since $\varepsilon \approx c(1 - \chi_4)$ ([N2], Prop. 4.3), this ratio tracks the next self-dual prolate rung, $(1 - \chi_8)/(1 - \chi_4) \asymp \lambda^8$, with a slowly drifting constant (≈ 0.2 at $\lambda = 2$, ≈ 0.4 at $\lambda = 4$). If $R(\tilde{k}_\lambda) - \varepsilon = O(\varepsilon)$ persists, as it does through $\lambda = 1.7$, the left side of Corollary 5.5 is of order $\log\lambda\,/\,(\mu_1/\varepsilon) \approx 10^{-8}$ at $\lambda = 4$ [C].

This reopens the prolate row of Proposition 3.1 in a sharper form. The operator-norm comparison recorded there is dead; the trial-vector comparison has polynomial room against a logarithmic requirement at every computed window. The analytic target it leaves is a two-rung prolate comparison for the Weil form:

$$R(\tilde{k}_\lambda) - \varepsilon(\lambda) \;\lesssim\; \lambda^A\,(1 - \chi_4(\lambda)), \qquad \mu_1(\lambda) \;\gtrsim\; \lambda^{-B}\,(1 - \chi_8(\lambda)), \qquad A + B < 8.$$

5.3 Where the content sits when the Hypothesis fails [P]

Theorem 5.6 (a finite exceptional set destroys normality). *Suppose ζ has at least one, and only finitely many, zeros off the critical line. Let $\lambda_j \to \infty$ with (a) holding for $QW^{N_j}_{\lambda_j}$, N_j large enough that $\varepsilon_{N_j}(\lambda_j) \to -\infty$, and $\widehat{\xi}_{\lambda_j}(0) \neq 0$. Then $S(\lambda_j) \to \infty$. In particular the family is not normal, and the convergence of Corollary 5.3 fails on every real segment.*

Proof. Step 1 ($\varepsilon \to -\infty$). Let $\rho_0 = \frac{1}{2} + \delta + it_0$ be an off-line zero of maximal displacement $\delta > 0$, and let $\Delta_1, \dots, \Delta_m$ be the finitely many nonzero frequencies $\pm(t' \mp t_0)$, $t' + t_0$ attached to the off-line zeros $\frac{1}{2} \pm \delta + it'$ of the same displacement, together with $2t_0$. Since the exponentials $e^{(\delta + i\Delta)v}$ are linearly independent, we can fix a real $\varphi \in C_c^\infty(-1, 1)$ with

$$\int \varphi(v) e^{\delta v}\, dv = 1, \qquad \int \varphi(v) e^{(\delta + i\Delta_j)v}\, dv = 0 \quad (1 \le j \le m).$$

Put $c = L/2 - 1$, $g(u) = \varphi(u - c)\cos(t_0 u + \frac{\pi}{2})$ and $f(u) = g(u) + g(-u)$. Then f is even and supported in the window, and by the explicit formula $QW_\lambda(f) = \sum_\rho \hat{f}(\gamma_\rho)^2$.

The on-line terms total at most $B = \sum_\gamma (|\widehat{\varphi}(\gamma - t_0)| + |\widehat{\varphi}(\gamma + t_0)|)^2 < \infty$, independent of λ, because $|\hat{f}|$ on $\mathbb{R}$ does not see the translation by c. In $\hat{f}(\gamma_0)$, $\gamma_0 = t_0 - i\delta$, the piece of f at the near edge contributes $O(e^{-\delta c})$. The piece at the far edge contributes $-\frac{i}{2} e^{\delta c}$, plus the conditions above, which kill every term carrying a factor $e^{i\Delta_j c}$. Hence the quadruple of ρ_0 contributes $4 \operatorname{Re} \hat{f}(\gamma_0)^2 = -e^{2\delta c}$, the other quadruples of displacement δ contribute nothing at order $e^{2\delta c}$, and those of smaller displacement contribute $o(e^{2\delta c})$.

Since $\|f\|$ is fixed, $\varepsilon(\lambda) \le QW_\lambda(f)/\|f\|^2 \le -C\lambda^{2\delta}$ for large λ. By density the same holds for $\varepsilon_N(\lambda)$ once N is large.

Step 2 (energy identity). For the ground state, $\sum_\rho \hat{\xi}_\lambda(\gamma_\rho)^2 = QW_\lambda(\xi_\lambda) = \varepsilon(\lambda)\|\xi_\lambda\|^2$. With $F_\lambda = \hat{\xi}_\lambda / \hat{\xi}_\lambda(0)$ and Plancherel,

$$\sum_\rho F_\lambda(\gamma_\rho)^2 = \frac{\varepsilon(\lambda)}{2\pi} \int_{\mathbb{R}} F_\lambda(t)^2\, dt.$$

Step 3. Suppose $S(\lambda_j) \le M$ along a subsequence. Then $a_1 \ge M^{-1/2}$, and Lemma 5.1(ii) gives $F_\lambda \ge e^{-1/3}$ on $|t| \le \tau = 1/(2\sqrt{M})$, so the right side is at most $\varepsilon(\lambda_j)\, \tau e^{-2/3}/\pi \to -\infty$. On the left, every on-line term is ≥ 0 because F_λ is real on $\mathbb{R}$, and the finitely many off-line terms are bounded below by $-\sum_{\text{off}} e^{2M|\gamma_\rho|^2}$ (Lemma 5.1(i)). This is a contradiction. □

Quantitatively, Step 3 forces $e^{2ST^2}\sqrt{S} \gtrsim \lambda^{2\delta}$ for a zero at height T, that is $S(\lambda) \gtrsim \delta \log\lambda / T^2$. This rate is weak; the experiment below shows that the actual growth is quadratic in $\log\lambda$.

5.4 The certificate on ζ [A]

`empirical/connes_route/probe_lp_transport_zeta.py` builds QW^N_λ with every entry in closed form — digamma and trigamma values, geometrically convergent tails, finite prime sums — which agrees with the quadrature-certified matrix of [N2] to 3×10^{-40} and is roughly seventy times faster. Precision is set to resolve $\varepsilon(\lambda)$, up to 140 digits, and $N \ge 2\lambda^2 L + 24$ resolves the height $2\pi\lambda^2$ at which the window's maximal real-zero density $L/2\pi$ meets the density of the zeta zeros.

λ	N	digits	ε	$\varepsilon_{\text{odd}}/\varepsilon$	μ_1/ε	$S(\lambda)$	$\lambda^2(S_\Xi - S)$	$\sup_{[0,1]}\lvert F_\lambda - \Xi/\Xi(0)\rvert$	real zeros	edge share	σ
1.3	25	40	2.44×10^{-2}	15	3.70×10^{1}	0.007820900258	1.0052	1.51×10^{-2}	25/25	—	0.6817

λ	N	digits	ε	$\varepsilon_{\text{odd}}/\varepsilon$	μ_1/ε	$S(\lambda)$	$\lambda^2(S_\Xi - S)$	$\sup_{[0,1]}\lvert F_\lambda - \Xi/\Xi(0)\rvert$	real zeros	edge share	σ
1.5	27	40	1.45×10^{-4}	92	1.89×10^{3}	0.012695014734		1.02×10^{-2}	27/27	—	0.4633
2.0	35	44	9.24×10^{-13}	588	1.90×10^{5}	0.017692899216		5.30×10^{-3}	35/35	0.174	0.2210
2.5	46	57	5.93×10^{-24}	1.53×10^{3}	1.32×10^{6}	0.019751739310		3.28×10^{-3}	46/46	2.9×10^{-3}	0.1412
3.0	63	74	3.69×10^{-38}	3.56×10^{3}	7.07×10^{6}	0.020812538906		2.24×10^{-3}	63/63	9.5×10^{-6}	0.1035
3.5	85	93	3.10×10^{-55}	7.45×10^{3}	3.06×10^{7}	0.021438145204		1.63×10^{-3}	85/85	6.4×10^{-9}	0.0820
4.0	112	116	4.01×10^{-75}	1.48×10^{4}	1.15×10^{8}	0.021837282903		1.24×10^{-3}	112/112	1.3×10^{-12}	0.0682
4.5	145	143	8.54×10^{-98}	2.29×10^{4}	2.93×10^{8}	0.022107498702		9.75×10^{-4}	145/145	4.1×10^{-17}	0.0586

Table 5.1 (the ζ ladder). Columns: - ε, μ_1: the smallest and second eigenvalues of QW_λ^N on even vectors; ε_{odd}: the odd ground. - S: from the moment formula. - sup column: $t \in [0,1]$. - real zeros: secular roots found real, out of N. - edge share: the fraction of $|\xi_\lambda|^2$ within $\frac{1}{2}$ of the window edges (a dash where the window is narrower than 1). - $\sigma = 24S/L^2$.

The first zero is $z_1 = 16.066$ and 14.175 at $\lambda = 1.3, 1.5$, and 14.13472514, which is γ_1 to ten digits, from $\lambda = 2$ on. N-stability: raising N by a quarter moves S by 4×10^{-6} (relative) at $\lambda = 4$.

Every rung satisfies (a), with the even–odd split resolved directly, so Observation 4.1's resolution claim is withdrawn. All N secular roots are real (the CvS mechanism), and S computed from the moment formula $S = L^2/24 + (L^2/2\pi^2)\sum_{k\geq 1} \xi_k/(k^2\xi_0)$ agrees to fifteen digits with $\sum z_j^{-2}$ over the zeros, including the frame zeros $s = k > N$ of the truncation. Three laws are visible: 1. **The scalar converges.** $S(\lambda) \uparrow S_\Xi = -\Xi''(0)/2\Xi(0) = 0.0231049931\ldots$ with $\lambda^2(S_\Xi - S(\lambda)) \to\approx 0.020$. The deficit is the contribution of the zeros above the Nyquist height $2\pi\lambda^2$, which the window cannot yet resolve. 2. **The interval converges.** $\sup_{[0,1]} |F_\lambda - \Xi/\Xi(0)|$ falls in step with $S_\Xi - S$, and $z_1(\lambda) = \gamma_1$ to ten digits from $\lambda = 2$ on. 3. **The ground state is a bulk state.** The share of $|\xi_\lambda|^2$ within $\frac{1}{2}$ of the window edges falls from 3×10^{-3} ($\lambda = 2.5$) to 10^{-9} ($\lambda = 3.5$) and below, and $\sigma(\lambda) := S(\lambda)/(L^2/24) \to 0$.

5.5 The matched pair: Davenport–Heilbronn through the same machine [A]

The Davenport–Heilbronn function $f(s) = \sum a_n n^{-s}$, $a = (0, 1, \kappa, -\kappa, -1)$ modulo 5, $\kappa = (\sqrt{10 - 2\sqrt{5}} - 2)/(\sqrt{5}-1)$, has real coefficients, the functional equation $\Lambda(s) = (5/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})f(s) = \Lambda(1-s)$, no Euler product, and zeros off the line. The certificate locates the first four at $0.808517 + 85.699348i$, $0.650830 + 114.163343i$, $0.574356 + 166.479306i$ and $0.724258 + 176.702461i$. `probe_lp_transport_dh.py` runs the identical code with three changes forced by the data: no pole terms, the archimedean kernel $e^{-y/2}$ in place of $e^{y/2}$ (from $\Gamma(\frac{s+1}{2})$), and $\log 5$ in the constant, with $\Lambda_f(n)$ on all n from $-f'/f$. The construction is validated three ways: Λ_f against $-f'/f(3)$; the archimedean functional against its frequency-space digamma form, to all digits; and the matrix entry $QW[V_0, V_0]$ against the zero side, summing all 75 zeros below height 130, the four off-line ones included, within the tail allowance.

[TABLE 5.2 here]

Observation 5.7 (the bulk–edge transition). *Below $\lambda \approx [\cdot]$ the DH ground state behaves exactly like zeta's: $\varepsilon > 0$ and exponentially small, (a) holds, the zeros of $\hat{\xi}_\lambda$ track the DH zeros on the line*

(5.0942, 8.9401, 12.137, ... at $\lambda = 3$), S rises toward $S_{\Xi_{DH}} = 0.0981$ with $\lambda^2(S_{\Xi_{DH}} - S) \approx 0.10 \approx 5 \times 0.020$ (the Nyquist law with conductor 5), and the edge share vanishes. Across the transition the form turns negative, one direction per resolved off-line quadruple; the ground state migrates to the window edges; and $\sigma(\lambda) \to 3$, which is exactly the value for $\cos(zL/2)$, the transform of a pair of edge spikes ($S = L^2/8$).

Observation 5.8 (where the content is not). *Hypothesis (a) holds for DH at every computed window past the onset, so its ground-state transforms have only real zeros there, while RH fails for DH. (a) fails only at the onset itself, where the first negative mode is odd.* Hence the equivalence "even ground $\iff$ RH" asserted in Observation 4.2 is false: evenness is the parity of the edge doublet, not the Hypothesis. In the matched pair the content sits in normality — the boundedness of S, equivalently the interval convergence of Corollary 5.3 — exactly as Theorem 5.6 predicts for a finite exceptional set.

Conjecture 5.9 (bulk–edge dichotomy). *Under (a), $\sigma(\lambda) \to 0$ if and only if the Riemann Hypothesis holds; if it fails, $\sigma(\lambda) \to 3$.* The "only if" direction for a finite exceptional set is Theorem 5.6, up to the rate. [C]

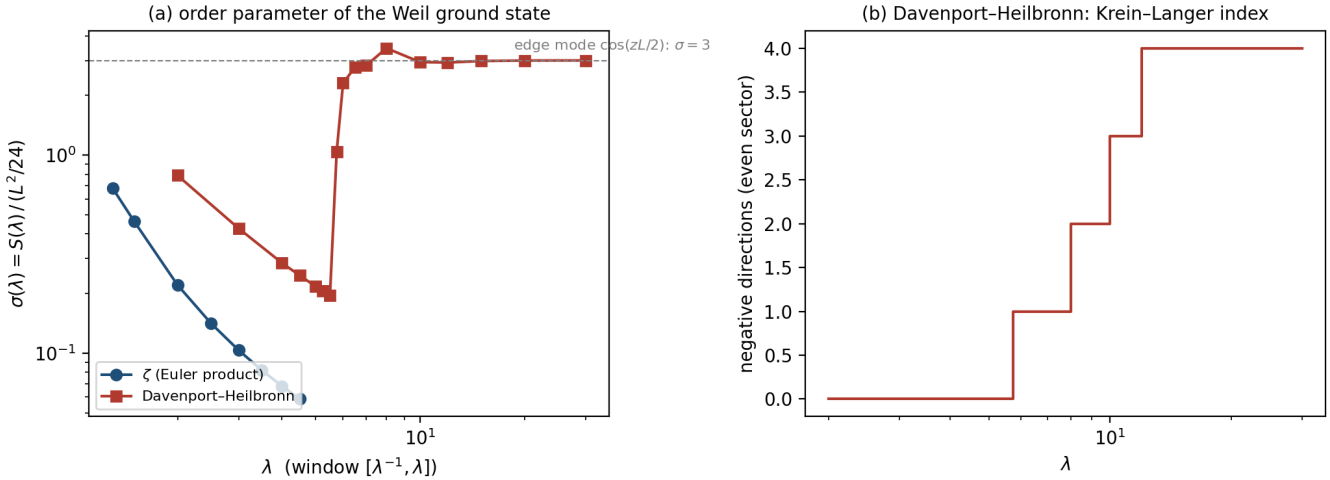


Figure 1: ...

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6. Honest status

What is **proved or certified** [P]: - the equivalence chain of §1; - the finite Weil form generating the zeta zeros on the line (Thm 2.1); - the prolate vehicle's simple even ground state (Thm 2.2); - the target identity $\widehat{\mathcal{E}(h)} = \Xi$; - the Laguerre–Pólya transport and its consequences for the open step (Theorem 5.2, Corollaries 5.3–5.5); - the destruction of normality by a finite exceptional set (Theorem 5.6).

What is **certified numerically** [A]: Table 5.1 for ζ and Table 5.2 for the Davenport–Heilbronn control, both computed from closed-form matrices at up to 143 digits.

What is **structural** [A]: the common closure point of six routes (Prop 3.1), which is $\varphi' \geq 0$, with the prolate row refined by Corollary 5.5.

What is **open** [C]: - the step restated in Section 4, namely (a) together with the interval convergence of Corollary 5.3, equivalently with the two-rung prolate comparison following Corollary 5.5; - the bulk–edge dichotomy (Conjecture 5.9).

The matched pair settles one point that had been open in the program's record. The evenness of the ground state is not where the Hypothesis lives; the normality of the approximant family is.

This program does not prove the Riemann Hypothesis, and N7 does not claim it. The reduction is exact, the finite half is realized, and the open step is now quantified. It is a convergence on one real interval, or equivalently a spectral inequality with a logarithmic margin, against observed margins of order λ^8 at every computed window. It remains open in the field's most advanced work (Connes–Consani–Moscovici, November 2025). The separation is the program's discipline: **structural reduction — real; proof of RH — not yet.**

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